

Mathematics 611: Algebraic Topology I

Fall 2021

Tuesdays, Thursdays 12:00–1:15 pm

Physics 227

Professor: Lenny Ng

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Office: Physics 216

Office hours (tentative): Tu 1:15–2:15, W 1–2

Sakai site: I'll be using Sakai for assignments and other information; please contact me if you need access and don't automatically have it. There is also a rudimentary public web page at <https://services.math.duke.edu/~ng/math611/>.

Course synopsis: Algebraic topology deals with the use of algebraic structures (such as groups, rings, and modules) to study and distinguish topological spaces. This course is essential background for research in topology and geometry as well as topological data analysis, and provides a framework that informs many other fields, including geometric analysis, number theory, and algebraic geometry.

Roughly speaking, we will cover Chapters 0, 1, and 2 of Hatcher. My plan is to cover the official Math 611 syllabus: <https://math.duke.edu/math-611-syllabus>. For your convenience, I'm reproducing that syllabus here:

- Homotopy and homotopy type, cell complexes, homotopy extension property.
- The fundamental group: homotopy of paths, homotopy-lifting property for paths through covering spaces, functoriality, Van Kampen's theorem, covering spaces, classification of covering spaces with the fundamental group, deck transformations and group actions, $K(G, 1)$ s.
- Singular homology, cellular homology, Mayer–Vietoris Sequences, homology with coefficients, relation with the fundamental group, relative homology, the long exact sequence of relative homology, applications to computing the homology of surfaces, projective spaces, axioms for homology, Euler characteristic.

Prerequisites: It is important that you are comfortable with both point-set topology (at the level of Math 411) and the algebra of groups and rings (at the level of Math 501). If you're not sure that you've taken these prerequisites or their equivalent, please consult with me to make sure that this course is appropriate for you.

Textbook: The textbook for this course is *Algebraic Topology* by Allen Hatcher. This is conveniently available for free online at <http://pi.math.cornell.edu/~hatcher/AT/ATpage.html>, though I recommend that you also purchase a physical copy of the book.

As supplementary texts, I recommend *Topology* and *Algebraic Topology* (that's two separate books) by James Munkres, and *A Basic Course in Algebraic Topology* by William Massey.

Office hours (subject to change): Tuesdays 1:15–2:15 and Wednesday 1:00–2:00, and by appointment (set up in person or by email).

Assignments: There will be weekly problem sets, which I expect to typically be due on Thursdays. Please check Sakai for posted problem sets. You can work with other students in the class on the homeworks, but please write up your problem sets on your own. My current plan is to ask you to submit your problem sets on paper; however, we might switch to Gradescope if this turns out to be more convenient.

Final exam: The final for this course will be a take-home final exam. You will have at least one week to work on the final, which will be due at 2 pm on December 13 (our final exam time slot).

Grading: Your grade is based on the problem sets and final exam.

Course duration: This class will run through the end of *undergraduate* courses, even though graduate courses ostensibly end a week earlier. This means that our last day of class is Thursday December 2.

Course notes: A version of my course notes for Math 611, from fall 2019, are available at <https://services.math.duke.edu/~ng/math611f19/>. Please note that they may contain errors, and they aren't a substitute for class attendance.

COVID considerations: This course meets in-person and I don't currently have plans to record or broadcast classes online. This may be subject to change if circumstances warrant. We will follow university guidelines and everyone should wear a mask during in-person classes and office hours.

It is crucial that you do not come to class if you have COVID symptoms or if you're asked by the university to quarantine. If you do need to miss class due to quarantine or isolation, please let me know as soon as possible and we can make arrangements; since there are lecture notes online, it should be fairly simple to make sure you miss as little as possible.

Here's the relevant language from the university: "Student health, safety, and well-being are the university's top priorities. To help ensure your well-being and the well-being of those around you, **please do not come to class** if you have symptoms related to COVID-19, have had a known exposure to COVID-19, or have tested positive for COVID-19. If any of these situations apply to you, you must follow university guidance related to the ongoing COVID-19 pandemic and current health and safety protocols. If you are experiencing any COVID-19 symptoms, contact student health, 919-681-9355. To keep the university community as safe and healthy as possible, you will be expected to follow these guidelines."